higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 3

(10041003)

14 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages.

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TIME: 3 HOURS
MARKS: 200

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 What is the most important element of an information system?

A Procedures
B People
C Machines
D Control

1.1.2 A company requires an information system to determine its ...

A payments or purchases.
B marketing or advertising.
C strengths or weaknesses
D None of the above-mentioned.

1.1.3 The type of processing that creates reports for supervisors and other functional users is called a …

A transaction processing system.
B management information system.
C executive support system.
D decision support system.

1.1.4 A program is not considered finished until …

A it has been used successfully for six months.
B the cost of the program breaks even with the benefit of the program.
C the ROI is positive.
D it has passed its testing.

1.1.5 Which of the following is NOT a factor in the failure of systems development projects?

A Size of the company
B Inadequate user involvement
C Failure of the integration of the system
D Continuation of a project that should have been cancelled

1.1.6 The starting point for developing an MIS is …

A identification of the business processes that are the essence of the business.
B distributed database management.
C the large number of PCs and the LAN of the system.
D the purchase of mini-computers.

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1.1.7 The person who ensures that systems are developed on time, within budget, and with acceptable quality is a …

A systems designer.
B project manager.
C systems developer.
D systems user.

1.1.8 When a bank offers web self-service to customers to answer their questions, the primary outcome is …

A adding value.
B managing risks.
C reducing costs.
D creating a new opportunity.

1.1.9 The general transformation cycle for information is …

A information to data to knowledge.
B knowledge to data to information.
C data to knowledge to information.
D data to information to knowledge.

1.1.10 Which one of the following is NOT a category of problems represented by the PIECES framework?

A Control
B Efficiency
C Service
D Technology

(10 × 1) (10)

1.2 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–O) next to the question number (1.1–1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	The transfer of information systems development, operation or maintenance to an outside firm that provides these services for a fee on a temporary or long-term basis	A	transaction processing system
1.2.2	A process by which business decisions are analysed	B	aim
1.2.3	The total value of the benefits minus the total value of the costs, with both costs and benefits adjusted to reflect the point in time at which they occur	C	request for proposal
1.2.4	A document that describes the company, lists the IT services or products needed, and specifies the features required	D	outsourcing
1.2.5	An assessment of the practicality of a proposed plan or method	E	discretionary
1.2.6	A company that develops software for sale	F	net present value
1.2.7	A percentage rate that compares the total net benefits (the return) received from a project to the total costs (the investment) of the project	G	vision
1.2.8	Process data generated by day-to-day business operations	H	return on investment
1.2.9	A name given to a systems development technique that tests systems concepts and provides an opportunity to examine input, output and user interfaces before final decisions are made	I	feasibility study
1.2.10	Projects that management have to implement	J	mission
		K	objective
		L	cost–benefit analysis
		M	prototyping
		N	goal
		O	software vendor

(10 × 1) (10)

1.3 Write the following abbreviations out in full:

1.3.1 RFP

1.3.2 MIS

1.3.3 RFQ

1.3.4 COO

1.3.5 CEO

1.3.6 TPS

1.2.7 OAS

1.3.8 POS

1.3.9 DBMS

1.3.10 EIS

(10 × 1) (10)

1.4 Choose the correct word from those given in brackets. Write only the word(s) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

1.4.1 (Ergonomics/Economics) is the study of health and comfort factors associated with users who spend a great deal of time working with computers at a desk.

1.4.2 The value of a (low-level/high-level) project schedule is that it provides important estimates regarding the resource requirements for a project.

1.4.3 (Resource management/Operational management) of information systems takes the view that technology, information systems staff and data are valuable resources that need to be appropriately managed, as they benefit the entire organisation.

1.4.4 (Attributes/Entities) are the actual facts or data which the business needs to know.

1.4.5 A (data store/process) represents data on a person, event, transaction or object which needs to be kept on record.

1.4.6 A (lease/purchase) is similar to a rental, but it is normally taken over a longer period and is used for high-value assets.

1.4.7 Leasing or purchasing is an example of (upgrading/financing) computer technology.

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1.4.8 (Strategic/Managerial) systems refer to those systems that support the planning and decision making of a business.

1.4.9 (The SDLC/A system) comprises a set of elements or things that are related or that interact to form a unified whole.

1.4.10 (Analysis/Design) determines what must be done to solve the problem.

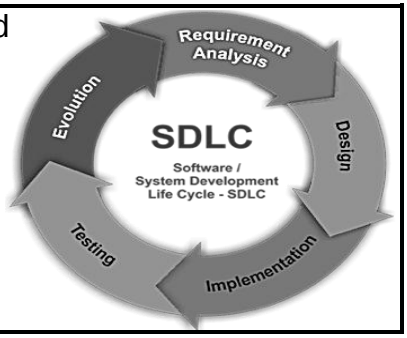
(10 × 1) (10)
[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

Systems development life cycle (SDLC), also referred to as the software development life cycle, is a term used in systems engineering, information systems and software engineering to describe a process for planning, creating, testing, and deploying an information system.



[Source: *https://en.wikipedia.org/wiki/Systems_development_life_cycle*]

2.1 What is an *information system*? (2)

2.2 List FOUR elements of a SDLC. (4)

2.3 List EIGHT possible causes of information systems development failures on initiated projects. (8)

2.4 Differentiate between the *phased development approach* and the *waterfall approach*. (4)
[18]

QUESTION 3

3.1 Define the term *data processing.* (2)

3.2 What is a *transaction processing system*? (3)

3.3 List THREE activities that are performed during data processing. (3)

3.4 Name TWO types of reports that will support operational decision making in management information systems. (2)

3.5 List and briefly explain THREE main categories of management support systems. (3 × 2) (6)
[16]

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QUESTION 4

> The information systems department is the formal organisational unit responsible for information technology services. The information systems department is responsible for maintaining the hardware, software, data storage, and networks that comprise the firm's IT infrastructure. The information systems department suggests new business strategies and new information-based products and services and coordinates both the development of the technology and the planned changes in the organisation.
>
> [Source: *https://paginas.fe.up.pt*]

4.1 Explain the roles of the following IT professionals in any modern organisation:

4.1.1 Chief knowledge officer

4.1.2 Webmaster

4.1.3 Chief information officer

4.1.4 Chief technology officer

4.1.5 Systems analyst

4.1.6 Database administrator

4.1.7 Computer programmer

4.1.8 Network administrator

4.1.9 Mobile app developer

(9 × 2) (18)

4.2 List FIVE tasks that are performed during information systems operations in an organisation. (5)

4.3 Differentiate between *operations support systems* and *systems support*. (4)

[27]

QUESTION 5

5.1 List and explain THREE methods of financing the acquisition of computer technology in an organisation. (3 × 2) (6)

5.2 List the SIX steps that should be taken when acquiring computer technology. (6)

5.3 Ongoing maintenance and care are important for the good performance and long life span of a PC.

Briefly explain how each of the following may affect a PC:

5.3.1 Heat

5.3.2 Dust

5.3.3 Magnets

5.3.4 Power surges

5.3.5 Electrostatics

(5 × 2) (10)

5.4 Differentiate between *information systems outsourcing* and *information systems offshoring*. (4)

[26]

QUESTION 6

> Decision making is the process whereby a course of action is selected from a number of alternatives. Like planning, decision making is all-pervasive and like forecasting, decision making forms an important part of planning. In any organisation, policy documents will assist managers in taking managerial decisions.

[Source: *www.yourarticlelibrary.com*]

6.1 What is an *organisation*? (2)

6.2 There are THREE main levels of decision making in an organisation.

List and explain the type of decisions that are taken by each of the following levels of management. Give ONE example of each.

6.2.1 Top level management

6.2.2 Middle level management

6.2.3 Lower/supervisory level management

(3 × 3) (9)

6.3 Define the term *positional analysis.* (3)

6.4 List FIVE positional analysis measures of the current state of an organisation or business. (5)

6.5 List FIVE competitive strategies that an organisation will formulate to maintain a competitive advantage and enhance organisational growth and expansion. (5)

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6.6 An information system planning is not done independently; it is done as part of organisational planning. The development of an information systems plan can be broken down into EIGHT steps.

List the EIGHT steps that are identified in the development of the information systems plan. (8)

6.7 Discuss FOUR categories of the BGC growth share matrix. (4 × 2) (8)

6.8 Draw any design or graphical representation that depicts Michael Porter's FIVE forces model for competitor analysis. (5 × 2) (10)

[50]

QUESTION 7

7.1 Define the following terms:

7.1.1 Cost–benefit analysis

7.1.2 Net present value

7.1.3 Request for proposal

(3 × 2) (6)

7.2 Depict the formula that is used to calculate the return on investment (ROI). (2)

7.3 Use the table below to calculate the return on investment on e-learning of the department of education.
Fill in the missing values or amounts from the data table (7.3.1–7.3.5) in the ANSWER BOOK.

Tasks	Total	Year 1	Year 2	Year 3	Year 4
Development costs	R1 000 000	R 1 000 000	-	-	-
Operational costs	**(7.3.1)**	R100 000	R120 000	R140 000	R160 000
Total costs	R1 520 000	**(7.3.2)**	R120 000	R140 000	**(7.3.3)**
Benefits – tangible	R1 800 000	R300 000	**(7.3.4)**	R 500 000	R600 000
Overall profit/losses	R280 000	**(7.3.5)**	R280 000	R360 000	R440 000

(5)

7.4 List the SIX steps that are involved in conducting and developing a cost–benefit analysis for an information systems development project. (6)

7.5 Differentiate between *project scoping* and *problem identification*. (4)

[23]

TOTAL SECTION B: 160
GRAND TOTAL: 200

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